Cross-Modality Contrastive Learning for Hyperspectral Image Classification

Renlong Hang, *Member, IEEE*, Xuwei Qian, and Qingshan Liu, *Senior Member, IEEE*

***Abstract*— Deep learning has attracted much attention in the field of hyperspectral image classification recently, due to its powerful representation and generalization abilities. Most of the current deep learning models are trained in a supervised manner, which requires large amounts of labeled samples to achieve the state-of-the-art performance. Unfortunately, pixel-level labeling in hyperspectral imageries is difficult, time-consuming, and human-dependent. To address this issue, we propose an unsupervised feature learning model using multimodal data, hyperspectral, and light detection and ranging (LiDAR) in particular. It takes advantage of the relationship between hyperspectral and LiDAR data to extract features, without using any label information. After that, we design a dual fine-tuning strategy to transfer the extracted features for hyperspectral image classification with small numbers of training samples. Such a strategy is able to explore not only the semantic information but also the intrinsic structure information of training samples. In order to test the performance of our proposed model, we conduct comprehensive experiments on three hyperspectral and LiDAR datasets. Experimental results show that our proposed model can achieve better performance than several state-of-the-art deep learning models.**

***Index Terms*— Fine-tuning strategy, hyperspectral image classification, multimodal data, unsupervised deep learning.**

I. Introduction

With the rapid development of imaging and spectroscopy technologies, nowadays, hyperspectral sensors are able to capture high-resolution imageries in both spectral and spatial domains. In hyperspectral imagery, each pixel records objects' detailed spectrum information with tens of or even hundreds of narrow and contiguous bands, while each spectral band, which consists of numerous pixels, describes spatial distributions of different objects in the observed scene. Such rich spectral–spatial information makes hyperspectral imageries different from multispectral and natural images, thus attracting more and more attention in the field of precision agriculture, climate monitoring, urban planning, and mineral exploitation, to name a few. For these applications, hyperspectral image classification is a common task, whose purpose is assigning one of predefined class labels to each pixel.

In order to achieve satisfactory classification performance, traditional models often focus on extracting handcrafted features [1]–[3] and/or designing advanced classifiers [4]–[6]. Recently, deep learning has revolutionized this pipeline [7], [8]. Different from the traditional models, it combines feature extraction and classifier design into a unified framework and optimizes them in a totally data-driven manner. Therefore, most researchers' efforts have changed to devising appropriate deep network structures [9]–[13]. For instance, based on convolutional neural networks (CNNs), Chen *et al.* [10] proposed three different architectures for learning spectral, spatial, and spectral–spatial features. These architectures demonstrated promising performance in comparison with traditional models, motivating lots of CNN-based models afterward. Nevertheless, most of the popular deep networks are supervised models, which often needs large amounts of training samples to learn. Visual interpretation and in-site survey are two basic approaches to label hyperspectral imageries, but both of them are expensive, time-consuming, and human-dependent. How to address the issue of limited training samples has become a key point for deep learning-based models.

In the past few years, a lot of works have targeted at dealing with the aforementioned issue. Among them, data augmentation, transfer learning, and semisupervised/unsupervised learning are three main kinds of solutions. Data augmentation is a pretty straightforward method, which attempts to enlarging the training set by taking advantage of available training samples. In [14], flipping and rotation operators were adopted to generate samples. Each training sample was first rotated by three different angles and then flipped along the vertical and horizontal directions. In [10], [15], and [16], random noise or occlusion was added to each training sample for creating new data. Qin *et al.* [17] applied a physical model named Hapke equations to simulate labeled hyperspectral data. Different from direct sample generation techniques mentioned above, there also exist other methods that aim to indirectly augment training samples [18]–[20]. For instance, a pixel-pair strategy was proposed in [18]. It aims to select any pairs of pixels from the training set and label them according to their original labels, thus dramatically increasing the number of training samples. Such pair strategy is often combined with metric learning [21]–[24], whose key idea is to pull intraclass samples closer and push interclass samples farther away.

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The authors are with the Engineering Research Center of Digital Forensics, Ministry of Education, the Jiangsu Collaborative Innovation Center of Atmospheric Environment and Equipment Technology, and the School of Computer and Software, Nanjing University of Information Science and Technology, Nanjing 210044, China (e-mail: renlong_hang@163.com; qxwturbo1996@163.com; qsliu@nuist.edu.cn).

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Except for exploiting the training set of the target hyperspectral imagery, one can also resort to auxiliary data whose labeling information already existed. Based on this idea, transfer learning has been developed and explored in the field of hyperspectral image classification [23], [25]. It often consists of two steps. The first step is pretraining a model on auxiliary data, which has enough numbers of training samples. The second step is fine-tuning the pretrained model on the target data. Considering the spectral configuration of different sensors, hyperspectral imageries collected from the same sensor as the target imagery were adopted as auxiliary data in [26] and [27]. Differently, some researchers argued that the knowledge learned from different hyperspectral sensors can also be transferred [28], [29]. In order to realize this goal, data preprocessing operators, such as dimensionality reduction [29], are desired to make consistent spectral dimensions between the auxiliary imagery and the target one. Jiao *et al.* [30] found that even the model trained with natural images is able to be transferred to hyperspectral image classification. In contrast with hyperspectral imageries, popular natural image datasets, such as ImageNet, contain millions of training samples, making it possible to take advantage of very deep networks. Similar ideas were proposed in [31] and [32]. However, due to the distribution gap between two different datasets, the performance of cross-scene transfer learning still has much room to improve. Domain adaptation [33]–[35] is a possible way to deal with this issue, which attempts at reducing the distribution difference in a shared space.

Although transfer learning is capable of decreasing the required number of training samples on target imageries, the labeling cost on the auxiliary data is nonnegligible. Actually, alongside with labeled samples, there exist significantly larger amounts of unlabeled samples, which are much cheaper to acquire. Unsupervised/semisupervised learning focuses on making full use of the rich information from them. Autoencoder (AE) is a typical category of unsupervised deep network [36]–[38]. Chen *et al.* [36] first applied it to the task of hyperspectral image classification. However, traditional AEs are developed to process vectorized data. In order to directly extract features from hyperspectral images, convolutional AEs (CAEs) were introduced in [39] and [40]. In recent years, a generative adversarial network (GAN) has been designed for unsupervised feature learning from hyperspectral imageries [41], [42]. The basic framework of GAN contains a generator and a discriminator. After the adversarial training between them, the optimized discriminator was used as an unsupervised feature extractor in [41] and [42]. Besides unsupervised learning, GAN can also be easily extended to semisupervised learning by incorporating the label information from training samples. For example, He *et al.* [43] and Feng *et al.* [44] modified the discriminator to not only discriminate "real" and "fake" samples but also classify the category of input samples; in [45], the generator was changed to reconstructing each input image patch and classifying the labeled training patches.

In this article, we put focus on unsupervised learning. Different from existing reconstruction- and GAN-related methods, our model targets at constructing an unsupervised deep learning framework by exploring multimodal information [46]. Over the same observation area, different types of sensors are able to record objects' information from diverse views. For example, hyperspectral provides reflectance information of objects, while light detection and ranging (LiDAR) describes their height information. Although these two kinds of information seem different, they should be similar to each other in some transformation space because both of them correspond to the same object. Inspired from this observation, we propose a cross-modality contrastive learning framework for unsupervised feature extraction. In order to effectively transfer the extracted features for downstream classification tasks, fine-tuning strategies are adopted, which often use limited training samples to optimize the classifier. In addition to the semantic information of training samples, here, we also explore their intrinsic structure information; thus, an enhanced classifier can be acquired. In summary, this article has the following contributions.

1) To address the issue of small numbers of training samples in the field of hyperspectral image classification, we propose an unsupervised learning framework. In contrast with most existing models, our framework is constructed upon paired multimodal data (i.e., hyperspectral and LiDAR) and a contrastive learning strategy.
2) In order to better transfer our framework to the classification task, we design a novel fine-tuning strategy, which not only takes advantage of the semantic information of each sample but also explores the intrinsic structure information between different samples.
3) Comprehensive experiments are conducted on three paired hyperspectral–LiDAR datasets, and the experimental results indicate that our proposed model is capable of achieving satisfactory classification performance with small numbers of training samples in each class.

The remaining sections are organized as follows. In Section II, the details of the proposed model are introduced. In Section III, experimental data and results are described. Finally, the conclusions of this article are summarized in Section IV.

II. Methodology

A. Architecture Overview

It is well acknowledged that the classification performance of hyperspectral imageries heavily depends on extracted or learned features. The majority of current state-of-the-art deep models often use large amounts of labeled samples to ensure the generalization ability of features. In order to alleviate the dependency of labeled samples, we propose an unsupervised feature learning model using multimodal data, which can extract well-generalized features, so that only a small number of labeled samples are enough to acquire high-performance classification results. Fig. 1 shows the flowchart of our proposed model, which mainly consists of two parts. The left part is the framework of unsupervised feature learning, which mainly takes advantage of cross-modality contrast information between hyperspectral and LiDAR data. The right part is a fine-tuning framework, which aims to transfer the learned

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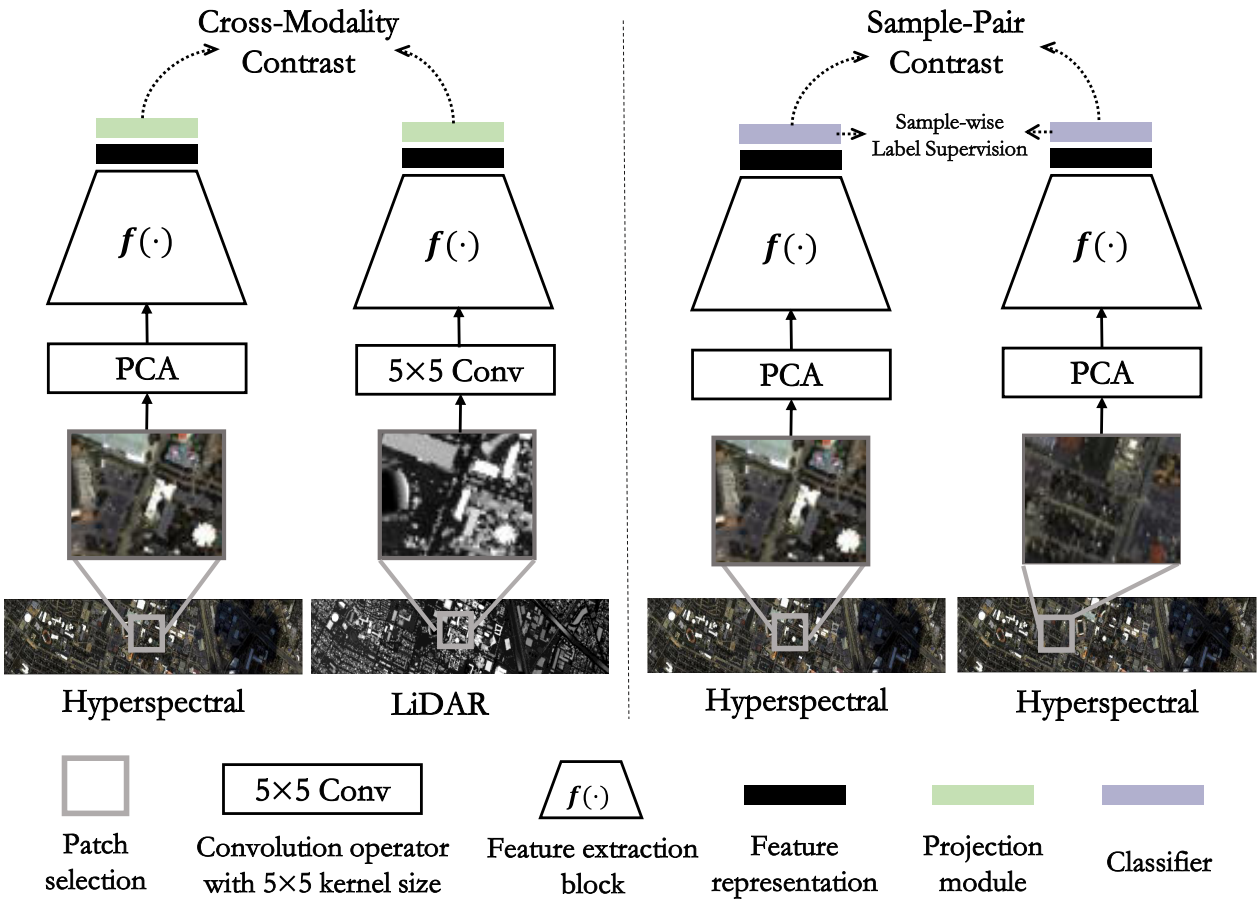


Fig. 1. Flowchart of the proposed model.

features from the left part to hyperspectral classification. In the following, we will introduce both parts in detail.

B. Unsupervised Feature Learning Framework

Assume that $\mathbf{X}_{\mathcal{H}} \in \Re^{w\times h\times b}$ and $\mathbf{X}_{\mathcal{L}} \in \Re^{w\times h}$ represent the hyperspectral and LiDAR data, respectively, where b is the number of spectral bands in the hyperspectral imagery and w and h, respectively, correspond to the width and height of both data. As shown in Fig. 1, we first select N patches with a fixed spatial size $s \times s$, which can be denoted as $x^i_{\mathcal{H}} \in \Re^{s\times s\times b}, i \in \{1, 2, \ldots, N\}$, from the hyperspectral imagery. At the same spatial positions, the corresponding N LiDAR patches, represented as $x^j_{\mathcal{L}} \in \Re^{s\times s}$, $j \in \{1, 2, \ldots, N\}$, are also chosen. For each $x^i_{\mathcal{H}}$, we employ principal component analysis (PCA), a simple yet effective dimensionality-reduction model, to reduce its spectral redundancy. Specifically, $x^i_{\mathcal{H}}$ needs to be first reshaped into a matrix, and then, PCA is adopted to reduce its dimensionality. Suppose that the number of preserved principal components is p. Meanwhile, a convolutional operator, whose kernel size is 5×5 and kernel number is $\tilde{p}$, is adopted to each $x^j_{\mathcal{L}}$. There exist two reasons for such operator. First, the original LiDAR data often contain some noise information in the spatial domain. The convolutional operator can be regarded as a spatial filter to extract robust and effective features. Second, the input channel number of $x^i_{\mathcal{H}}$ and $x^j_{\mathcal{L}}$ is different. The convolutional operator is able to make them consistent by setting $\tilde{p} = p$. After that, both hyperspectral and LiDAR patches are fed into a shared feature extraction block $f(\cdot)$ to learn feature representations $z^i_{\mathcal{H}}$ and $z^j_{\mathcal{L}}$. Fig. 2 shows the detailed structure of $f(\cdot)$, which is comprised of three successive convolution blocks. Each block contains a 1×1 convolution layer to extract high-dimensional features, a 3×3 convolution layer to strengthen the high-dimensional representation, and a 2×2 average pooling layer to increase the receptive field. Note that there is a residual connection in the 3×3 convolution layer, which enhances the information reuse in the network. From the first block to the third block, the number of kernels in each block is sequentially set to 128, 256, and 512, respectively.

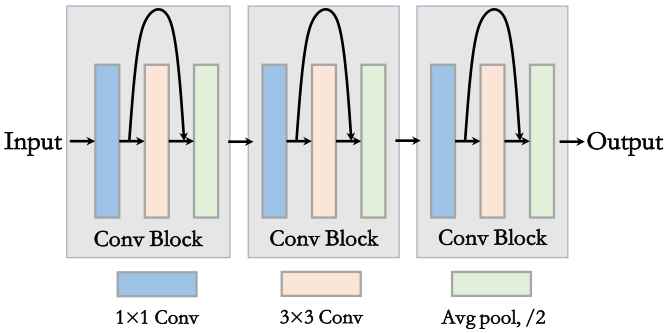


Fig. 2. Structure of the feature extraction block in Fig. 1.

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Following the feature extraction block, a shared projection module is designed to map $z^i_{\mathcal{H}}$ and $z^j_{\mathcal{L}}$ into a hidden space. This module is composed of two fully connected layers, and the first one of them is followed by a batch normalization (BN) layer and a ReLU nonlinear activation layer. The whole process can be formulated as

$$\begin{cases} h^i_{\mathcal{H}} = w_2^\top \left[\mathrm{ReLU}\left(\mathrm{BN}\left(w_1^\top z^i_{\mathcal{H}}\right)\right)\right] \\ h^j_{\mathcal{L}} = w_2^\top \left[\mathrm{ReLU}\left(\mathrm{BN}\left(w_1^\top z^j_{\mathcal{L}}\right)\right)\right] \end{cases}$$

where $h^i_{\mathcal{H}}$ and $h^j_{\mathcal{L}}$ are the projection results of $z^i_{\mathcal{H}}$ and $z^j_{\mathcal{L}}$, respectively. In the equation, w_1 represents the connection weight between the feature extraction block and the first fully connected layer which contains 512 neurons. Similarly, w_2 denotes the connection weight between the first fully connected layer and the second one containing 16 neurons. In the hidden space, $h^i_{\mathcal{H}}$ and $h^j_{\mathcal{L}}$ are expected to be much closer when i and j point to the same spatial position than other situations. Inspired from [47], we propose a cross-modality contrastive loss to achieve this goal, which is formulated as

$$\ell_{i,j} = -\log \frac{\exp\left(\mathrm{sim}\left(h^i_{\mathcal{H}}, h^j_{\mathcal{L}}\right)/\tau\right)}{\Pi} \tag{1}$$

where $\ell_{i,j}$ represents the loss function for the paired $x^i_{\mathcal{H}}$ and $x^j_{\mathcal{L}}$, Π equals $\sum_{k=1}^{N} \mathbf{1}_{[k\neq j]} \exp(\mathrm{sim}(h^i_{\mathcal{H}}, h^k_{\mathcal{L}})/\tau) + \mathbf{1}_{[k\neq i]} \exp(\mathrm{sim}(h^i_{\mathcal{H}}, h^k_{\mathcal{H}})/\tau)$, $\mathbf{1}_{[k\neq j]} \in \{0, 1\}$ is an indicator function whose value equals 1 if $k \neq j$, and τ is a hyperparameter, which is empirically set to 1 in this article. The function $\mathrm{sim}(\cdot, \cdot)$ computes the cosine similarity between input vectors. For example, $\mathrm{sim}(h_{\mathcal{H}}, h_{\mathcal{L}}) = h_{\mathcal{H}}^\top h_{\mathcal{L}} / \|h_{\mathcal{H}}\| \|h_{\mathcal{L}}\|$, where $\|\cdot\|$ denotes the modulus of a vector. During the training process, each paired hyperspectral and LiDAR data in a mini-batch needs to compute their loss values according to (1). These values are summarized together to update the network parameters. It is worth noting that the paired hyperspectral and LiDAR data here mean that both hyperspectral and LiDAR patches are cropped from the same spatial position but different modalities. Since the whole process does not rely on any label information of hyperspectral or LiDAR data, it can be regarded as an unsupervised feature learning framework.

C. Feature Fine-Tuning for Hyperspectral Classification

After unsupervised pretraining, another critical issue is how to transfer it for the classification task, where a limited number of labeled samples from the hyperspectral imagery are available. Fine-tuning is a widely adopted strategy to address this issue. It often inputs the learned representation into a classifier and use the label information to update the network parameters in a supervised manner. We can also follow this pipeline for our application. Specifically, we directly reuse the left hyperspectral branch and replace the projection module by a classifier. Assume that $\{x^i_{\mathcal{H}}, y^i\}_{i=1}^{M}$ represents the training set, where y^i is the label of $x^i_{\mathcal{H}}$ and M is the total number of training samples. Then, we apply a cross-entropy loss $\mathcal{L}_{\mathrm{CE}}$ to update network parameters, which can be denoted as

$$\mathcal{L}_{\mathrm{CE}} = -\sum_{i=1}^{M} y^i \log \frac{\exp\left(w_i^\top z^i_{\mathcal{H}}\right)}{\sum_{j=1}^{C} \exp\left(w_j^\top z^i_{\mathcal{H}}\right)} \tag{2}$$

where C is the number of predefined categories to classify and w_i is the weight for the ith class output in the classifier.

As shown in (2), $\mathcal{L}_{\mathrm{CE}}$ mainly explores the semantic information of each sample to fine-tune the network parameters. In addition to it, there also exist some structural relationships between different samples. For example, intraclass samples are much closer than interclass ones in the same space. Considering that (1) is able to describe the pairwise relations, it is natural to extend it for constructing a loss function to capture such structural relationships. For a pair of training samples from the same class, we assume that they should be close to each other, so the loss function $\mathcal{L}_{\mathrm{Stru}}$ can be constructed as

$$\mathcal{L}_{\mathrm{Stru}} = -\sum_{i=1}^{M} \log \frac{\sum_{j=1}^{M} \mathbf{1}_{[y^i=y^j]} \exp(\mathrm{sim}(\tilde{y}_i, \tilde{y}_j)/\tau)}{\sum_{k=1}^{M} \mathbf{1}_{[y^k\neq y^i]} \exp(\mathrm{sim}(\tilde{y}_i, \tilde{y}_k)/\tau)} \tag{3}$$

where $\tilde{y}_i$ is the predicted results for $x^i_{\mathcal{H}}$. The final loss function is a weighted combination of $\mathcal{L}_{\mathrm{CE}}$ and $\mathcal{L}_{\mathrm{Stru}}$

$$\mathcal{L} = \mathcal{L}_{\mathrm{CE}} + \lambda \mathcal{L}_{\mathrm{Stru}} \tag{4}$$

where λ is a hyperparameter to balance these two loss functions. Given a training batch, (2) is first employed to calculate the loss for each sample. Then, a lot of sample pairs are generated by any two samples, and (3) is applied to compute another loss. These two losses are finally combined with (4), which is capable of exploring not only samplewise semantic information but also the sample pair relationships. Note that in the fine-tuning stage, LiDAR data are no longer needed.

D. Model Training and Testing

For a given hyperspectral–LiDAR dataset, three subsets are needed to be constructed, including a training set, a candidate set, and a test set. Both the candidate set and the test set without label information can be used to train the unsupervised feature learning framework, as shown in the left of Fig. 1. The training set is adopted to train the fine-tuning framework for the classification task, as shown in the right of Fig. 1. The test set is used to evaluate the classification performance of the fine-tuned model. To reduce the effects of random sampling, five repeated experiments are conducted and the averaged performance is reported in terms of the overall accuracy (OA), the average accuracy (AA), the per-class accuracy, and the Kappa coefficient.

Our models are implemented in the PyTorch framework and Adam is chosen as the optimizer. During the unsupervised feature learning stage, we adopt a large batch size (i.e., 2048). The learning rate and the training epochs are chosen as 0.001 and 200, respectively. At the classification stage, the batch size and the learning rate are changed to 32 and 0.0002, respectively. All the experiments are deployed on a personal computer with an Intel core i5-9400F, 2.90-GHz processor, 16-GB RAM, and an RTX 2080Ti graphic card.

III. Experiments

A. Data Description

In order to evaluate the performance of our proposed model, we conduct experiments on three widely used hyperspectral–LiDAR datasets.

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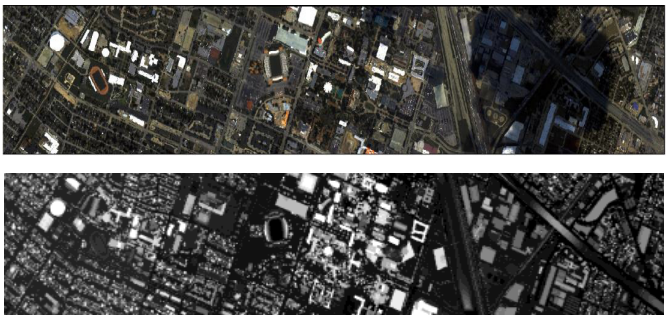


Fig. 3. Visualization of the Houston dataset. (Top) Pseudo-color image of the hyperspectral data. (Bottom) Gray image for the LiDAR data.

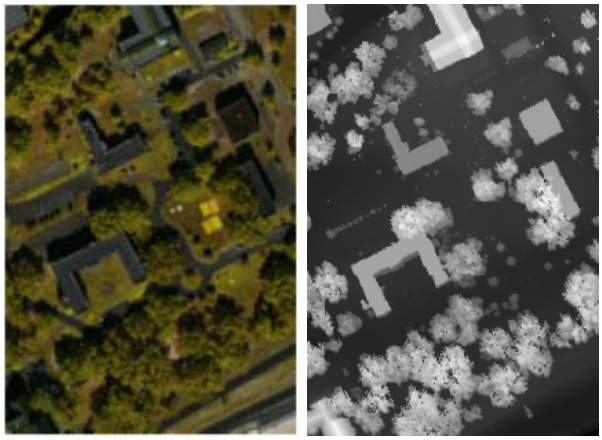


Fig. 4. Visualization of the MUUFL dataset. (Left) Pseudo-color image of the hyperspectral data. (Right) Gray image for the LiDAR data.

The first one is the Houston dataset [48]. It was acquired on June 2012, over the University of Houston campus and the neighboring area. Both hyperspectral and LiDAR data have a spatial size of 349 × 1905 pixels, containing 15 land cover classes. The spatial resolution of them is 2.5 m. For the hyperspectral data, there exist 144 spectral bands with the wavelength ranging from 0.38 to 1.05 μm. Fig. 3 shows a pseudo-color image of the hyperspectral data and a gray image of the LiDAR data. In Table I, the number of available samples in different categories is reported. Note that the effect of cloud shadows in the hyperspectral data was detected and eliminated with respect to the assumption that cloud shadows are larger than structures on the ground.[1]

The second one is the Muufl Gulfport (MUUFL) dataset [49], acquired over the University of Southern Mississippi Gulf Park Campus on November 8, 2010. It is comprised of coregistered hyperspectral and LiDAR data, both of which have a spatial resolution of 1 m. The hyperspectral data, consisting of 64 spectral bands, were collected using the CASI-1500 sensor. The spatial size of both hyperspectral and LiDAR data is 325 × 220 pixels. Fig. 4 visualizes a pseudo-color image of the hyperspectral data and a gray image of the LiDAR data. There are 11 classes of land covers on this dataset, and the number of samples in each class is shown in Table II.

[1]The enhanced dataset was provided by Prof. Naoto Yokoya from RIKEN, Saitama, Japan.

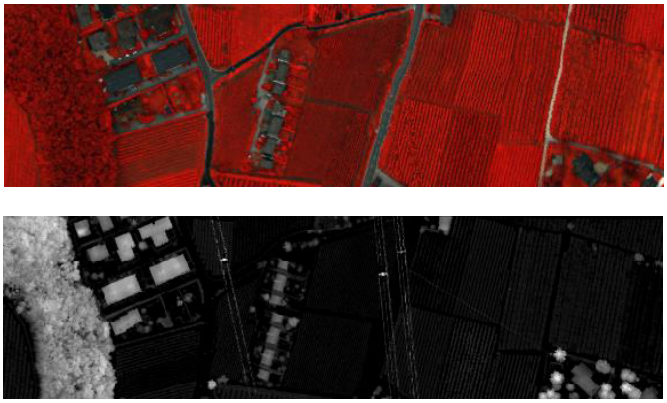


Fig. 5. Visualization of the Trento dataset. (Top) Pseudo-color image of the hyperspectral data. (Bottom) Gray image for the LiDAR data.

TABLE I

SAMPLE DISTRIBUTIONS ON THE HOUSTON DATASET

Class No.	Class Name	Training	Candidate	Test
1	Healthy grass	10	198	1053
2	Stressed grass	10	190	1064
3	Synthetic grass	10	192	505
4	Tree	10	188	1056
5	Soil	10	186	1056
6	Water	10	182	143
7	Residential	10	196	1072
8	Commercial	10	191	1053
9	Road	10	193	1059
10	Highway	10	191	1036
11	Railway	10	181	1054
12	Parking lot 1	10	192	1041
13	Parking lot 2	10	184	285
14	Tennis court	10	181	247
15	Running track	10	187	473
-	Total	150	2832	12197

The third one is the Trento dataset. It was collected over a rural area in the south of Trento, Italy, with the AISA Eagle sensor for the hyperspectral data and the Optech ALTM 3100EA sensor for the LiDAR data. The spatial resolution and size for both data are 1 m and 166 × 600 pixels, respectively. The number of spectral bands for the hyperspectral data is 63, covering the wavelength from 0.42 to 0.99 μm. Fig. 5 shows a pseudo-color image of the hyperspectral data and a gray image of the LiDAR data. There exist six categories of land covers to recognize, and the available samples in each class are reported in Table III.

B. Hyperparameter Selection

There are three hyperparameters in our proposed model. They are the input patch size $s \times s$, the preserved number of principal components p in PCA, and the regularization parameter λ in (4). In this section, we attempt to analyze their effects on classification performance and select the optimal value for each one.

1) Analysis on s: To evaluate the effects of s, we choose different values from a candidate set $\{9, 11, 13, 15, 17, 19\}$ and test the classification performance on three datasets. Fig. 6 shows the OA values achieved by our proposed model

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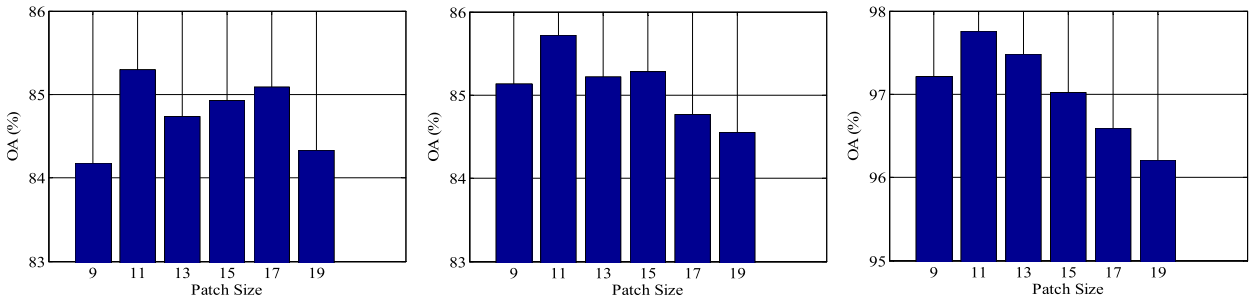


Fig. 6. Classification performance in terms of OA using different s values on three datasets. The Houston, the MUUFL, and the Trento dataset are from left to right.

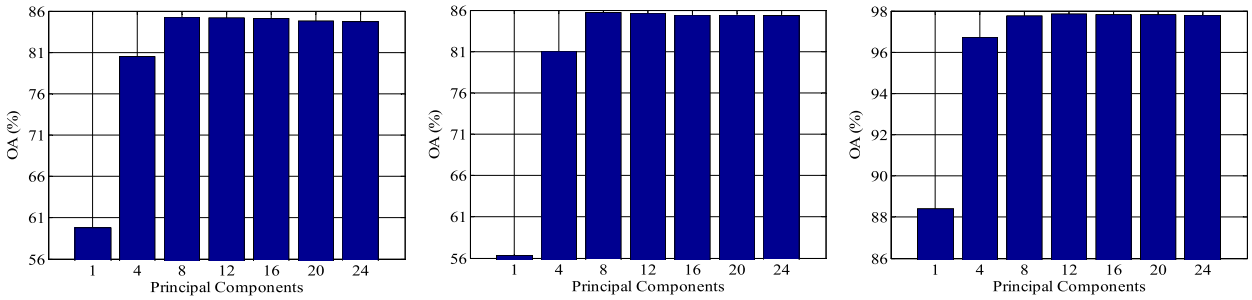


Fig. 7. Classification performance in terms of OA using different p values on three datasets. The Houston, the MUUFL, and the Trento dataset are from left to right.

TABLE II
SAMPLE DISTRIBUTIONS ON THE MUUFL DATASET

Class No.	Class Name	Training	Candidate	Test
1	Trees	10	150	23096
2	Mostly grass	10	150	4120
3	Mixed ground surface	10	150	6732
4	Dirt and sand	10	150	1676
5	Road	10	150	6537
6	Water	10	150	316
7	Building shadow	10	150	2083
8	Building	10	150	6090
9	Sidewalk	10	150	1235
10	Yellow curb	10	150	33
11	Cloth panels	10	150	119
-	Total	110	1650	52037

TABLE III
SAMPLE DISTRIBUTIONS ON THE TRENTO DATASET

Class No.	Class Name	Training	Candidate	Test
1	Apple trees	10	129	3905
2	Buildings	10	125	2778
3	Ground	10	105	374
4	Wood	10	154	8969
5	Vineyard	10	184	10317
6	Roads	10	122	3052
-	Total	60	819	29395

using different patch sizes. It can be observed as s changes from 9 to 11, and OA is dramatically improved on three datasets. However, when s is larger than 11, the changing trend is different between Houston and the other two datasets. In particular, OAs achieved on Houston data first increase from 13 to 17 and then decrease from 17 to 19. Differently, on MUUFL and Trento, OAs gradually decrease as s increases. For all three datasets, they have a consistent optimal s value (i.e., 11) since the highest OA appears at such point.

2) Analysis on p: Similar to the analysis on s, we fix the other hyperparameter values and change p values from a candidate set $\{1, 4, 8, 12, 16, 20, 24\}$. The detailed classification results in terms of OA are shown in Fig. 7. When only one principal component is used, OAs are very low on three datasets because a lot of discriminative information is lost. As p value increases from 1 to 4, OAs are improved by more than 20% on Houston and MUUFL and about 8% on Trento. When we further increase p, OA still improves, but not so much as that from 1 to 4. The highest OA values can be obtained when p equals 8 on all datasets. Therefore, the optimal p value is set to 8.

3) Analysis on λ: The same as above processes, we fix s and p to their respective optimal values and then select λ values from a set $\{0.01, 0.1, 1, 10, 100\}$. Fig. 8 shows OAs using different λ values. It is interesting to observe that OAs are gradually increasing when λ values are less than 10. Specifically, OA is changed from about 83.5% to more than 85% on the Houston dataset, from 84% to more than 85.5% on the MUUFL dataset, and from about 96% to more than 97.5% on the Trento dataset. However, there exists a significant decrease from 10 to 100 on all datasets. Based on such observations, it is clear to set λ as 10.

C. Model Comparison

In order to validate the effectiveness of our proposed model, we compare it with several state-of-the-art models, including a traditional unsupervised model named PCA_RF, two supervised deep models named CNN and EMFNet, a semisupervised deep model named MTGAN, and four

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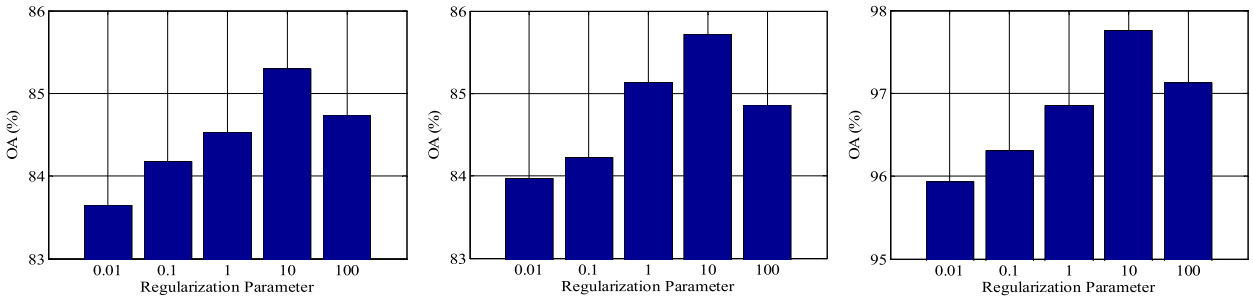


Fig. 8. Classification performance in terms of OA using different λ values on three datasets. The Houston, the MUUFL, and the Trento dataset are from left to right.

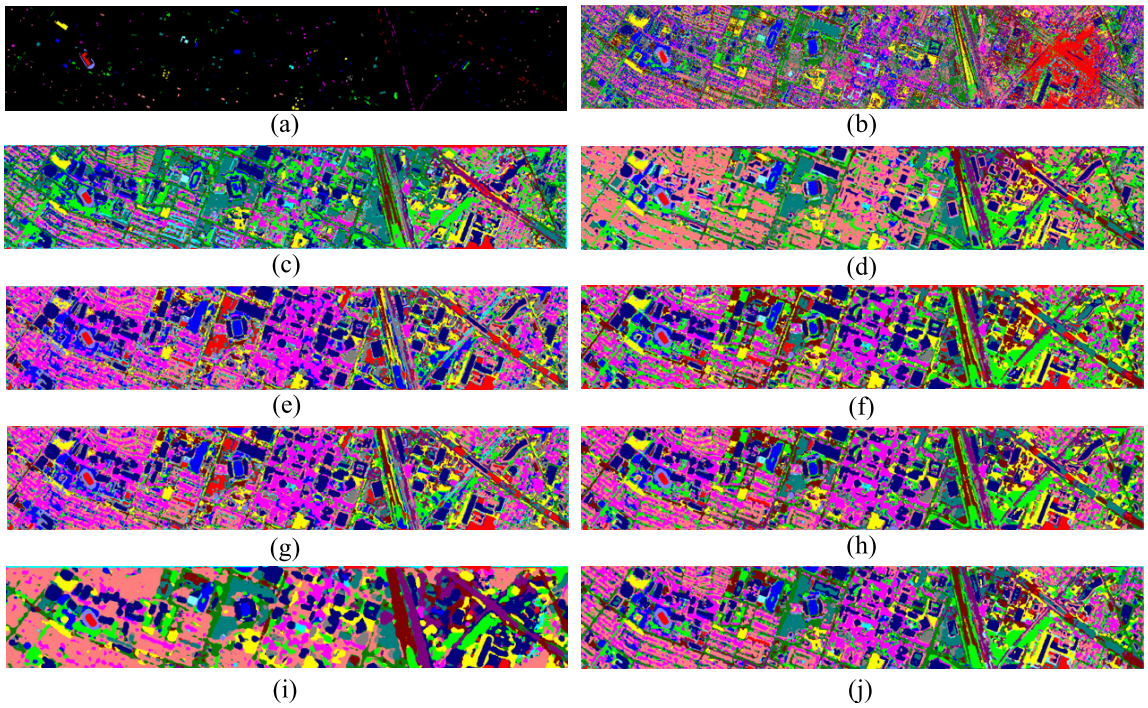


Fig. 9. Classification maps achieved by different models on the Houston dataset. (a) Test data map. (b) PCA_RF. (c) CNN. (d) EMFNet. (e) AE_H. (f) CAE_H. (g) AE_HL. (h) CAE_HL. (i) MTGAN. (j) Ours.

unsupervised deep models, including AE_H, CAE_H, AE_HL, and CAE_HL. Some explanations about them are given as follows.

1) *PCA_RF:* It first employs PCA to extract spectral features from hyperspectral data and then adopts random forest with ten trees to classify the extracted features.
2) *CNN:* It is comprised of three convolutional layers with 3×3 kernels and a fully connected layer to classify hyperspectral patches. The kernel numbers from the first to the third convolutional layer are 32, 64, and 128, respectively. The input patch size is 11×11.
3) *EMFNet:* This network was designed to fuse hyperspectral and LiDAR data in [50]. We adopt the same framework to classify hyperspectral and LiDAR data.
4) *MTGAN:* It is a GAN-based semisupervised model proposed in [45]. The generator of this model is designed to reconstruct and classify input patches simultaneously. We adopt the same framework as in [45].
5) *AE_H:* It consists of an encoder and a decoder, both of which contain three fully connected layers. The input of the encoder is a vectorized hyperspectral patch with a size of 11×11.
6) *CAE_H:* Different from AE_H, this network directly uses hyperspectral patches as inputs. Three convolutional layers and three deconvolutional layers are used to construct the network.
7) *AE_HL:* It adopts the same network structure as AE_H but replaces the input by concatenated hyperspectral and LiDAR patches.
8) *CAE_HL:* Similar to CAE_H, it applies the same number of convolutional and deconvolutional layers. However, the inputs are concatenated hyperspectral and LiDAR patches.

The quantitative comparison results on three datasets are reported in Tables IV–VI, where the bold font in each row corresponds to the best result. Several conclusions can be observed from them. For unsupervised feature extraction models, AE_H achieves inferior performance in terms of OA, AA, and Kappa than PCA_RF on the Houston and the MUUFL data. Although AE_HL significantly improves the performance

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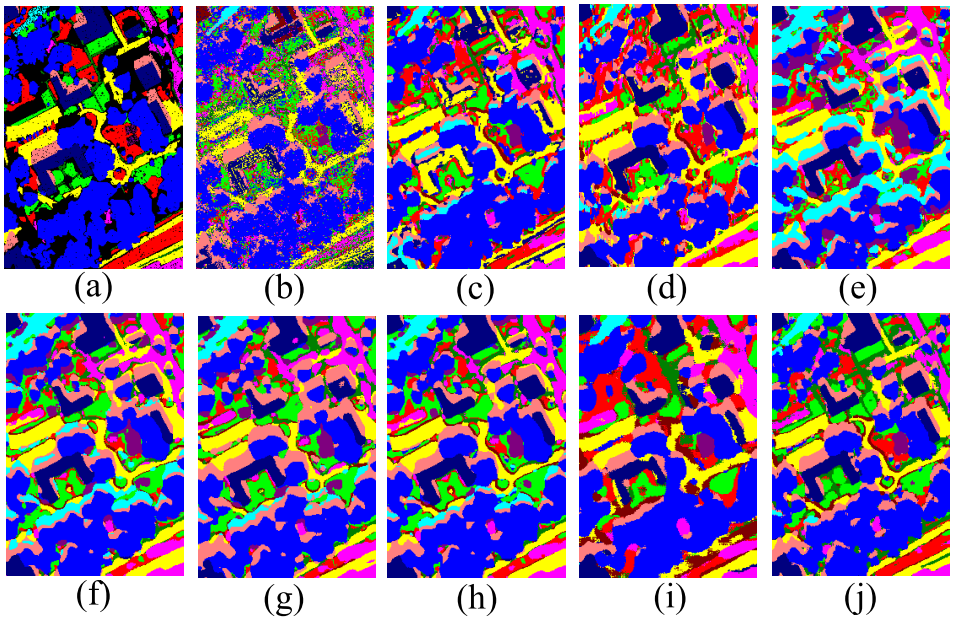


Fig. 10. Classification maps achieved by different models on the MUUFL dataset. (a) Test data map. (b) PCA_RF. (c) CNN. (d) EMFNet. (e) AE_H. (f) CAE_H. (g) AE_HL. (h) CAE_HL. (i) MTGAN. (j) Ours.

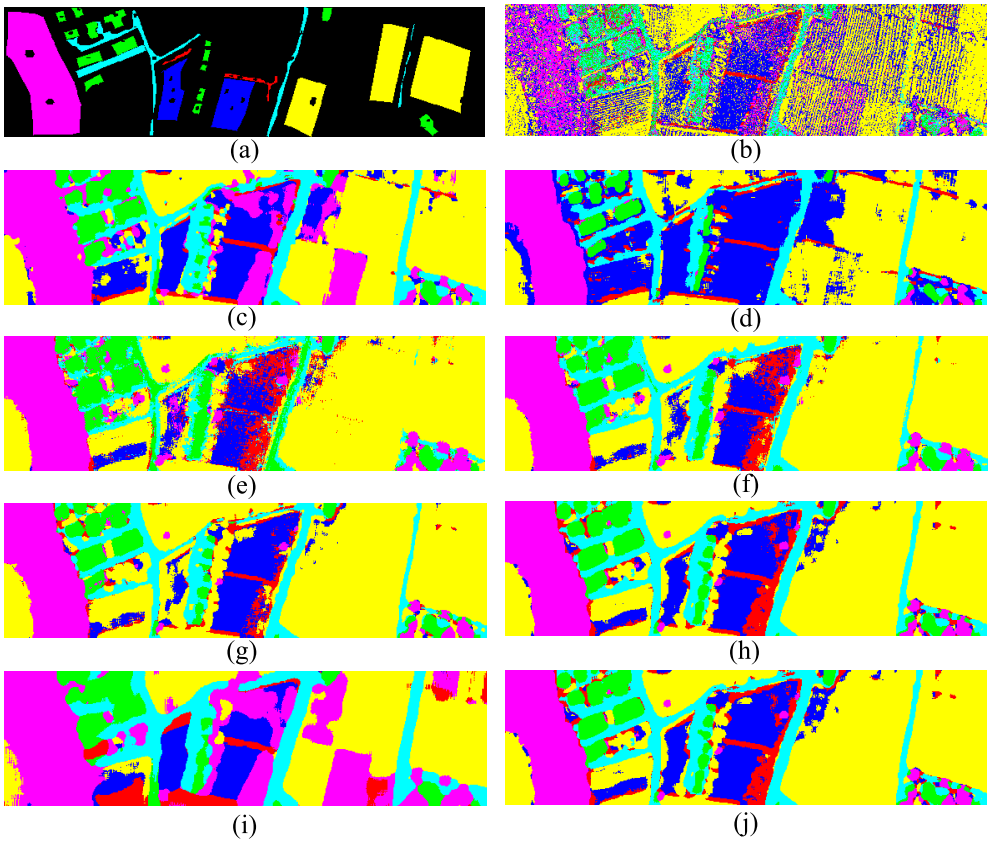


Fig. 11. Classification maps achieved by different models on the Trento dataset. (a) Test data map. (b) PCA_RF. (c) CNN. (d) EMFNet. (e) AE_H. (f) CAE_H. (g) AE_HL. (h) CAE_HL. (i) MTGAN. (j) Ours.

of AE_H by incorporating the LiDAR information, its performance is still not better than PCA_RF on the first two datasets. However, on the Trento data, both AE_H and AE_HL have superior performance in terms of OA, AA, and Kappa when compared with PCA_RF. It indicates that AE, whose primary purpose is data reconstruction, is not always obtaining discriminative features as expected. Different from AE, CAE-based models get much higher OA, AA, and Kappa values

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TABLE IV
CLASSIFICATION PERFORMANCE OF DIFFERENT MODELS ON THE HOUSTON DATASET

Class No.	PCA_RF	CNN	EMFNet	AE_H	CAE_H	AE_HL	CAE_HL	MTGAN	Ours
1	79.15	77.18	79.68	**95.31**	82.94	89.44	81.99	77.53	83.08
2	77.52	95.99	87.69	8.97	74.61	47.35	92.03	**98.57**	96.00
3	83.41	88.55	95.05	**95.84**	94.85	95.76	92.32	95.37	95.01
4	80.23	97.50	79.43	98.13	99.22	98.73	**99.41**	88.43	98.84
5	82.67	96.59	95.15	83.92	94.79	95.30	99.81	98.18	**100**
6	91.19	81.68	64.34	19.58	79.02	61.68	85.17	93.85	**97.20**
7	63.38	58.12	80.04	72.85	81.47	85.28	89.48	83.23	**93.28**
8	35.78	57.38	78.77	83.74	76.05	83.63	86.82	71.28	**88.13**
9	59.00	70.33	60.30	23.97	33.28	30.22	44.04	72.28	**80.49**
10	28.98	27.34	44.81	26.27	47.70	36.66	53.15	42.80	**57.55**
11	47.63	72.24	**84.63**	54.27	62.69	66.28	83.32	82.94	79.05
12	40.44	**72.64**	71.03	9.414	28.11	14.52	37.10	69.70	52.60
13	23.44	85.61	81.54	**94.04**	86.53	92.07	88.42	90.18	92.49
14	80.89	82.02	87.45	8.99	52.47	15.22	66.72	**94.09**	84.78
15	94.12	68.50	88.29	89.89	98.60	95.48	**99.70**	81.14	98.73
OA	61.87	73.72	77.68	58.27	70.71	66.92	78.52	80.15	**84.55**
AA	64.52	75.45	78.55	57.68	72.82	67.17	79.97	82.64	**86.48**
Kappa	0.5879	0.7151	0.7583	0.5494	0.6828	0.6422	0.7671	0.7845	**0.8324**

TABLE V
CLASSIFICATION PERFORMANCE OF DIFFERENT MODELS ON THE MUUFL DATASET

Class No.	PCA_RF	CNN	EMFNet	AE_H	CAE_H	AE_HL	CAE_HL	MTGAN	Ours
1	77.01	**88.31**	84.60	59.95	75.28	73.31	83.96	87.58	85.03
2	55.85	67.81	67.39	34.06	63.51	62.79	74.81	68.79	**84.08**
3	51.94	45.32	42.18	39.12	32.20	23.32	30.89	49.99	**56.60**
4	70.26	74.03	69.99	85.63	86.07	83.85	**89.43**	78.28	84.22
5	67.22	73.61	73.29	78.71	80.93	68.96	83.84	74.87	**87.58**
6	79.30	85.51	82.41	99.49	**100**	99.56	**100**	87.53	99.87
7	78.84	66.96	74.82	42.44	78.72	86.65	90.95	81.15	**97.84**
8	54.06	55.11	85.49	84.49	89.88	83.26	93.05	70.08	**94.10**
9	43.21	43.72	32.96	10.96	24.86	22.35	34.48	28.31	**74.28**
10	76.36	58.18	57.58	9.70	36.36	26.06	47.27	55.15	**82.42**
11	96.47	93.45	81.34	**99.16**	**99.16**	**99.16**	**99.16**	93.61	98.99
OA	67.29	72.99	74.31	59.69	70.66	66.48	76.81	75.62	**83.01**
AA	68.23	68.36	68.37	58.52	69.73	66.30	75.26	70.49	**85.91**
Kappa	0.5829	0.6477	0.6675	0.5159	0.6337	0.5841	0.7034	0.6833	**0.7818**

TABLE VI
CLASSIFICATION PERFORMANCE OF DIFFERENT MODELS ON THE TRENTO DATASET

Class No.	PCA_RF	CNN	EMFNet	AE_H	CAE_H	AE_HL	CAE_HL	MTGAN	Ours
1	61.21	99.36	**99.49**	67.34	72.61	72.98	92.90	97.31	94.83
2	56.33	88.76	95.73	98.42	**99.70**	96.93	98.45	89.37	99.18
3	73.10	73.05	71.12	30.86	55.45	60.16	72.67	87.06	**87.86**
4	68.83	99.39	99.15	94.11	**100**	99.09	99.53	99.78	99.44
5	59.57	99.85	99.04	**100**	**100**	**100**	**100**	99.28	99.99
6	36.23	69.50	80.73	39.81	74.08	78.08	**83.14**	77.04	82.61
OA	60.06	95.11	96.56	86.58	93.07	93.06	96.67	95.77	**97.10**
AA	59.21	88.32	90.88	71.76	83.64	84.54	91.11	91.64	**93.99**
Kappa	0.4864	0.9343	0.9539	0.8186	0.9062	0.9057	0.9552	0.9434	**0.9611**

than PCA_RF- and AE-based models. Taking Houston and MUUFL datasets as instances, OA, AA, and Kappa values are increased by more than 10% from AE_HL to CAE_HL. These observations show that the spatial information is of great importance for the classification task.

For supervised models, EMFNet is capable of achieving better results than CNN on all datasets, due to the full exploitation of complementary information between hyperspectral and LiDAR data. When we compare the best unsupervised model CAE_HL with the best supervised model EMFNet, it is interesting to see that CAE_HL is able to acquire higher classification performance than EMFNet, especially on the MUUFL dataset. A similar conclusion can be observed from MTGAN. It achieves better classification results than EMFNet in terms of OA, AA, and Kappa values on the first two datasets. These conclusions demonstrate that taking advantage of unlabeled samples is beneficial once the available training samples are small. In comparison with CAE_HL and MTGAN, our proposed model can further improve the classification performance on all datasets. Specifically, OA, AA, and Kappa

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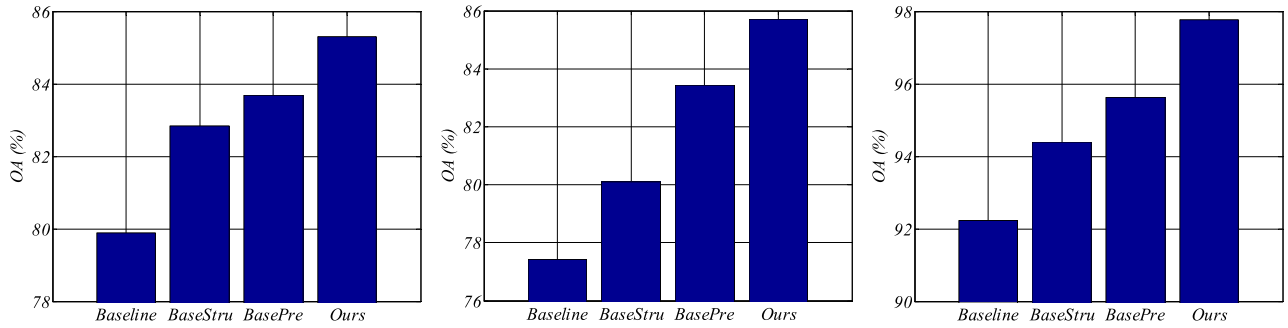


Fig. 12. Classification performance using different components on three datasets. The Houston, the MUUFL, and the Trento dataset are from left to right.

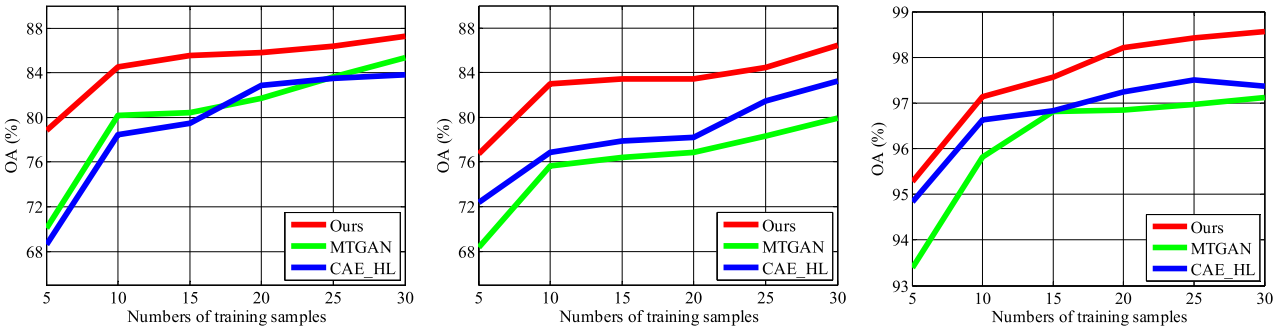


Fig. 13. Classification performance using different numbers of training samples on three datasets. The Houston, the MUUFL, and the Trento dataset are from left to right.

are improved by more than 4% on the first two datasets. Different from the Houston and MUUFL datasets, the Trento dataset is much easier to classify. Therefore, most of the comparison models achieve more than 90% OA values, and CAE_HL makes a very high baseline (i.e., 96.67%). Nevertheless, our proposed model still improves OA and Kappa values to 97.10% and 0.9611, respectively.

In terms of each class accuracy, our proposed model obtains the best results in more classes than comparison models on the Houston and MUUFL datasets. Although CAE_H gets higher accuracies than our model when classifying *buildings*, *wood*, and *vineyard* on the Trento dataset, such improvements are not high. In contrast, our model dramatically improves the classification accuracies on the other three classes (i.e., *apple trees*, *ground*, and *roads*), making AA increasing from 83.64% to 93.99%. Besides the quantitative results, Figs. 9–11 qualitatively show the classification maps of different models on three datasets. Compared to the ground-truth map, our model achieves more consistent results than the other models. All the above observations can sufficiently certify the effectiveness of our proposed model.

D. Ablation Study

Our proposed model includes unsupervised pretraining, supervised fine-tuning with a cross-entropy loss $\mathcal{L}_{\text{CE}}$, and a structural relationship-based loss $\mathcal{L}_{\text{Stru}}$. To test the effects of each component, we conduct ablation experiments in this section. In particular, three models are constructed except for the proposed model. First, we use the classification network in the right of Fig. 1 to construct a baseline model and train it with $\mathcal{L}_{\text{CE}}$. For simplicity, we name this model as baseline. Then, we replace the loss function with a weighted combination of $\mathcal{L}_{\text{CE}}$ and $\mathcal{L}_{\text{Stru}}$, as described in (4). This model is referred to as BaseStru. Finally, we add unsupervised pretraining to baseline and build a model named BasePre. Fig. 12 shows the classification performance of them in terms of OA on three datasets. It can be observed that BaseStru improves OAs by about 2% compared to baseline, indicating the effectiveness of our proposed loss $\mathcal{L}_{\text{Stru}}$. In comparison with BaseStru, BasePre further boosts OA values, which certifies the superiority of our pretraining framework. More interestingly, when all components are combined together, our proposed model achieves the highest OA values. Based on the aforementioned observations, we can conclude that each component is important in our proposed model.

E. Effects of Training Sample Numbers

All the aforementioned results are based on a fixed number of training samples. It would also be interesting to test the robustness of our proposed model with different numbers of training samples. To this end, we adopt 5–30 training samples in each class for experiments and compare our proposed model with MTGAN and CAE_HL because both of them can obtain relatively better classification results than the other comparison models. Fig. 13 shows the OA values achieved by three different models with different training sample numbers. It can be observed that as the number of training samples increases from 5 to 30, the OA values improve significantly for all models, which indicates the importance of training sample numbers for deep learning models. In addition, using the same number of training samples, CAE_HL is able to achieve higher OA values than MTGAN in most cases. In comparison with

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CAE_HL, our proposed model shows superior performance on all three datasets, which certifies the robustness and effectiveness of it.

IV. Conclusion

In this article, we proposed an unsupervised feature learning framework using multimodal data (i.e., hyperspectral and LiDAR). In contrast with the widely used reconstruction-based frameworks, our proposed model aims at exploring cross-modality information. Given hyperspectral and LiDAR patches cropped from the same spatial position, our model first extracted features from both modal data independently, and then, a shared projection module was devised to map the extracted features into a common hidden space. Considering that different-modal patches correspond to the same object when they lie in the same position, their feature representations should be close to each other in some spaces. To this end, a cross-modality contrastive loss was designed. At the classification phase, a dual fine-tuning strategy was proposed to exploit both the semantic and the structural information from small numbers of training samples. In order to test the performance of our proposed model, three hyperspectral–LiDAR datasets were adopted. Experimental results show that our model is able to achieve better performance than several state-of-the-art models. In addition, hyperparameter analysis and ablation study were conducted to evaluate the effects of different hyperparameter values and components on classification performance.

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Renlong Hang (Member, IEEE) received the Ph.D. degree from the School of Computer and Software, Nanjing University of Information Science and Technology, Nanjing, China, in 2017.

From 2018 to 2019, he was a Post-Doctoral Researcher with the Department of Computer Science and Electrical Engineering, University of Missouri–Kansas City, Kansas City, MO, USA. He is currently an Associate Professor with the School of Computer and Software, Nanjing University of Information Science and Technology. He has authored or coauthored over 30 peer-reviewed articles in international journals, such as IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, and IEEE GEOSCIENCE AND REMOTE SENSING LETTERS. His research interests include machine learning, pattern recognition, and their applications to remote sensing image processing.



Xuwei Qian received the B.E. degree in electric automation from the Chengxian College, Southeast University, Nanjing, China, in 2018, and the M.S. degree in control engineering from the Nanjing University of Information Science and Technology, Nanjing, in 2022.

His research interests include the efficient training and inference of deep learning models.



Qingshan Liu (Senior Member, IEEE) received the M.S. degree from Southeast University, Nanjing, China, in 2000, and the Ph.D. degree from the Chinese Academy of Sciences, Beijing, China, in 2003.

From 2010 to 2011, he was an Assistant Research Professor with the Department of Computer Science, Computational Biomedicine Imaging and Modeling Center, Rutgers, The State University of New Jersey, Piscataway, NJ, USA. Before he joined Rutgers University, he was an Associate Professor with the National Laboratory of Pattern Recognition, Chinese Academy of Sciences, Beijing, China. From June 2004 to April 2005, he was an Associate Researcher with the Multimedia Laboratory, The Chinese University of Hong Kong, Hong Kong. He is currently a Professor with the School of Computer and Software, Nanjing University of Information Science and Technology, Nanjing. His research interests include image and vision analysis.

Dr. Liu received the President Scholarship of the Chinese Academy of Sciences in 2003.

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